Solutions – Practice Problem Set 1

ECON 320 – Introduction to Mathematical Economics

Fall 2025

1. **Solve inequalities and express in interval notation.**

   (a) $|2x - 3| \le 5$.
   By definition, $-5 \le 2x - 3 \le 5$. Add 3: $-2 \le 2x \le 8$. Divide by 2: $\boxed{-1 \le x \le 4}$, i.e. $\boxed{[-1, 4]}$.

   (b) $(x - 3)(x + 2) \le 0$.
   Roots at $x = -2, 3$. Since it is a product of two linear factors, it is ≤ 0 *between* the roots. Hence $\boxed{[-2, 3]}$.

   (c) $\dfrac{1}{x-2} \ge 3$.
   Case $x > 2$: multiply by $x - 2 > 0$ gives $1 \ge 3(x - 2) \iff 3x \le 7 \iff x \le \frac{7}{3}$.
   Together with $x > 2$, we get $(2, \frac{7}{3}]$.
   Case $x < 2$: multiply by $x - 2 < 0$ *reverses* inequality: $1 \le 3(x - 2) \implies x \ge \frac{7}{3}$, impossible with $x < 2$.
   Therefore $\boxed{(2,\ 7/3]}$.

2. **Set operations for $A = [-1, 4)$ and $B = (1, 6]$.**

   (a) $A \cup B$ spans from the least left endpoint to the greatest right endpoint without gaps: $\boxed{[-1, 6]}$.
   $A \cap B$ is the overlap: $(1, 4)$. So $\boxed{A \cap B = (1, 4)}$.

   (b) $A \setminus B$: points in A not in $B \Rightarrow \boxed{[-1, 1]}$.
   $B \setminus A$: points in B not in $A \Rightarrow \boxed{[4, 6]}$.

   (c) For $A \cap B = (1, 4)$, $\inf = 1$, $\sup = 4$. Hence $\boxed{\inf = 1,\ \sup = 4}$.

3. **Domain of** $f(x) = \dfrac{\sqrt{5 - \ln(x^2 - 9)}}{x - 3}$.
   Conditions: (i) $\ln(x^2 - 9)$ defined $\Rightarrow x^2 - 9 > 0 \Rightarrow |x| > 3$; (ii) radicand $\ge 0 \Rightarrow 5 - \ln(x^2 - 9) \ge 0 \iff \ln(x^2 - 9) \le 5 \iff x^2 - 9 \le e^5 \iff |x| \le \sqrt{9 + e^5}$; (iii)

denominator $\neq 0$ (already enforced by $x > 3$ with $x \neq 3$).
Combine: $\boxed{\left[-\sqrt{9+e^5},\,-3\right) \,\cup\, \left(3,\,\sqrt{9+e^5}\right]}$.

4. **Inverse of** $f(x) = \dfrac{2x-3}{x+1}$ **(domain** $x > -1$**); evaluate** $f^{-1}(5)$**.**
Solve $y = \dfrac{2x-3}{x+1} \Rightarrow y(x+1) = 2x-3 \Rightarrow x(y-2) = -(y+3) \Rightarrow \boxed{f^{-1}(y) = \dfrac{y+3}{2-y}}$.
Since $x > -1$, $f(x) = 2 - \dfrac{5}{x+1}$ has range $(-\infty, 2)$. Therefore $f^{-1}(5)$ **is undefined** (because $5 \notin (-\infty, 2)$). The algebraic value $-8/3$ lies outside the domain $x > -1$.

5. **Solve for** x**.**
   (a) $e^{3x} = 7 \Rightarrow 3x = \ln 7 \Rightarrow \boxed{x = \tfrac{1}{3}\ln 7}$.
   (b) $2^{x+1} = 5e^{0.2x} \Rightarrow (x+1)\ln 2 = \ln 5 + 0.2x$.
   $x(\ln 2 - 0.2) = \ln(5/2) \Rightarrow \boxed{x = \dfrac{\ln(5/2)}{\ln 2 - 0.2}} \approx \boxed{1.858}$.

6. **Sequence** $S = \left\{\dfrac{2n+1}{n+3} : n \in \mathbb{N}\right\}$ (take $\mathbb{N} = \{1, 2, \ldots\}$).
Rewrite: $\dfrac{2n+1}{n+3} = 2 - \dfrac{5}{n+3}$. Thus $\lim_{n\to\infty} = 2$. The sequence is increasing and bounded above by 2, with smallest term at $n = 1$: $3/4$. Hence $\boxed{\inf S = 3/4,\ \sup S = 2}$ and $\boxed{\lim S = 2}$.

7. **Set** $B = \{(x, y) \in \mathbb{R}^2_{\geq 0} : 2x + 3y \leq 60\}$.
Boundary $2x + 3y = 60$ gives intercepts: $x = 30$ (when $y = 0$), $y = 20$ (when $x = 0$). So the feasible set is a right triangle. Area $= \frac{1}{2}(30)(20) = \boxed{300}$. In function form: $y \leq \frac{60-2x}{3}$ for $0 \leq x \leq 30$. Hence $\boxed{B = \{(x, y) : 0 \leq x \leq 30,\ 0 \leq y \leq (60-2x)/3\}}$.

8. **Composition and transformation.** $u(x) = \ln x$, $g(x) = 2 - 3\ln(4-x)$.
Domain: $4 - x > 0 \Rightarrow \boxed{x < 4}$. Composition: $\boxed{g(x) = 2 - 3\,u(4-x)}$. Interpretation: start with $u(x)$, reflect horizontally across $x = 2$ (since $x \mapsto 4-x$), then scale vertically by 3 and reflect (minus sign), and shift up by 2.

9. **Solve** $|x - a| \leq b$ $(b > 0)$.
By definition, $-b \leq x - a \leq b \Rightarrow a - b \leq x \leq a + b$. Hence $\boxed{[a-b,\ a+b]}$.

10. **Domain of** $h(x) = \sqrt{9-x^2} + \dfrac{1}{\sqrt{x-1}}$.
Require $9 - x^2 \geq 0 \Rightarrow -3 \leq x \leq 3$, and $x - 1 > 0 \Rightarrow x > 1$ (strict because of denominator). Intersection: $\boxed{(1, 3]}$.

11. **Differentiate and evaluate.**

(a) $y = \dfrac{(3x^2+1)^5}{xe^{2x}}$. Use logarithmic differentiation:

$$\ln y = 5\ln\big(3x^2+1\big) - \ln x - 2x \;\Rightarrow\; \frac{y'}{y} = \frac{30x}{3x^2+1} - \frac{1}{x} - 2.$$

At $x=1$: $y(1) = \dfrac{4^5}{e^2} = \dfrac{1024}{e^2}$ and $\frac{y'}{y}\big|_1 = \frac{30}{4} - 1 - 2 = \frac{9}{2}$. Hence

$$\boxed{y'(1) = \frac{1024}{e^2}\cdot\frac{9}{2} = \frac{4608}{e^2}}.$$

(b) $y = \ln\big(\sqrt{1+4x^2}\big) = \frac{1}{2}\ln(1+4x^2)$. Then

$$y' = \tfrac{1}{2}\cdot\frac{8x}{1+4x^2} = \boxed{\frac{4x}{1+4x^2}}, \qquad y'' = \frac{(4)(1+4x^2) - 4x(8x)}{(1+4x^2)^2} = \boxed{\frac{4-16x^2}{(1+4x^2)^2}}.$$

12. **Cost & average cost.** $C(q) = 0.5q^2 + 10q + 400$.

$MC = C'(q) = \boxed{q+10}$, $AC = C/q = \boxed{0.5q + 10 + \dfrac{400}{q}}$.

At $q=20$: $MC = 30$, $AC = 10 + 10 + 20 = \boxed{40}$.

Minimize AC: $AC'(q) = 0.5 - 400/q^2 = 0 \Rightarrow q^2 = 800 \Rightarrow \hat{q} = \boxed{20\sqrt{2}}$. Then

$$AC(\hat{q}) = 0.5(20\sqrt{2}) + 10 + \frac{400}{20\sqrt{2}} = 10\sqrt{2} + 10 + 10\sqrt{2} = \boxed{20\sqrt{2}+10}\ (\approx 38.284).$$

13. **Elasticity of demand.**

(a) $Q(P) = 1000P^{-1.2}$. Then $dQ/dP = -1.2\cdot 1000P^{-2.2}$. Elasticity $\varepsilon = \frac{dQ}{dP}\frac{P}{Q} = \boxed{-1.2}$ (constant).

(b) $Q(P) = 200 - 2P$. Then $\varepsilon = \frac{-2P}{200-2P}$. At $P=50$, $Q=100$, so $\boxed{\varepsilon = -1}$ (unit elastic).

14. **Implicit differentiation.** $xe^x + y^2 = 10$. Differentiate:

$$e^x(x+1) + 2y\,y' = 0 \;\Rightarrow\; \boxed{y' = -\frac{e^x(x+1)}{2y}}.$$

At $x=1$: $e + y^2 = 10 \Rightarrow y = \sqrt{10-e}$ (take $y>0$). Thus $\boxed{y'(1) = -\dfrac{e}{\sqrt{10-e}}}$.

15. **Differentials / percentage changes.** $Q = K^{0.3}L^{0.7}P^{0.1}$. For small changes,

$$\frac{\mathrm{d}Q}{Q} \approx 0.3\frac{\mathrm{d}K}{K} + 0.7\frac{\mathrm{d}L}{L} + 0.1\frac{\mathrm{d}P}{P}.$$

With $+5\%$, -2%, $+10\%$: $\Delta Q/Q \approx 0.3(0.05) + 0.7(-0.02) + 0.1(0.10) = \boxed{0.011} = \boxed{+1.1\%}$.

16. **Tax t in linear market.**

$$Q_s = -20 + 4(P - t), \quad Q_d = 700 - 6P.$$

Equate: $-20 + 4(P - t) = 700 - 6P \Rightarrow 10P = 720 + 4t \Rightarrow \boxed{P(t) = 72 + 0.4t}$. Then

$$Q(t) = 700 - 6P = 700 - 6(72 + 0.4t) = \boxed{268 - 2.4t}.$$

Hence $\boxed{\frac{\mathrm{d}P}{\mathrm{d}t} = 0.4}$, $\boxed{\frac{\mathrm{d}Q}{\mathrm{d}t} = -2.4}$. Interpretation: buyers bear 40% of the marginal tax (price to them rises by \$0.40 per \$1 tax).

17. **Profit maximization & comparative statics.** $\pi(q) = pq - (aq + \frac{b}{2}q^2) = (p-a)q - \frac{b}{2}q^2$.

FOC: $(p-a) - bq = 0 \Rightarrow \boxed{q^* = \frac{p-a}{b}}$ (since $b > 0$). Then $\boxed{\frac{\partial q^*}{\partial p} = 1/b > 0}$, $\boxed{\frac{\partial q^*}{\partial a} = -1/b < 0}$.

18. **Elasticity of supply.** $Q(P) = cP^\eta \Rightarrow \frac{\mathrm{d}Q}{\mathrm{d}P} = c\eta P^{\eta-1}$. Then

$$\varepsilon_s = \frac{\mathrm{d}Q}{\mathrm{d}P}\frac{P}{Q} = \frac{c\eta P^{\eta-1}P}{cP^\eta} = \boxed{\eta}.$$

For $c = 5$, $\eta = 0.8$: $\boxed{\varepsilon_s = 0.8}$ at any P (constant).

19. **Implicit-function comparative statics.** $F(x, a) = x^3 - 4x + a = 0$. By the IFT,

$$\frac{\mathrm{d}x}{\mathrm{d}a} = -\frac{F_a}{F_x} = -\frac{1}{3x^2 - 4}.$$

At $x = 2$: $\boxed{\mathrm{d}x/\mathrm{d}a = -1/8}$.

20. **Dynamic comparative statics.**

$$\begin{cases} \alpha Y - \beta P = Q, \\ \gamma P - \delta W = Q. \end{cases}$$

Solve: $\alpha Y - \beta P = \gamma P - \delta W \Rightarrow (\beta + \gamma)P = \alpha Y + \delta W$, hence

$$\boxed{P(Y, W) = \frac{\alpha Y + \delta W}{\beta + \gamma}}, \qquad \boxed{Q(Y, W) = \frac{\gamma\alpha}{\beta + \gamma}Y - \frac{\beta\delta}{\beta + \gamma}W}.$$

Differentiate w.r.t. time:

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{\alpha}{\beta + \gamma}\dot{Y} + \frac{\delta}{\beta + \gamma}\dot{W}}, \qquad \boxed{\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{\gamma\alpha}{\beta + \gamma}\dot{Y} - \frac{\beta\delta}{\beta + \gamma}\dot{W}}.$$

With $\alpha = \beta = \gamma = \delta = 2$: $\frac{\mathrm{d}Q}{\mathrm{d}t} = \dot{Y} - \dot{W}$. Given $\dot{Y} = 3$, $\dot{W} = -1$, obtain $\boxed{\mathrm{d}Q/\mathrm{d}t = 4}$.

21. **Log-linear approximation for profits.**
$R(q) = Aq^\theta \Rightarrow MR = A\theta q^{\theta-1}$, $C(q) = c_0 + c_1 q \Rightarrow MC = c_1$. For a small change Δq near q_0,

$$\Delta\pi \approx (MR_0 - MC_0)\,\Delta q = \left(A\theta q_0^{\theta-1} - c_1\right)(q_0\ \underbrace{\Delta q/q_0}_{\%\,\text{change}}).$$

Numerically, $A = 200$, $\theta = 0.6$, $c_1 = 20$, $q_0 = 50$, $+2\%$ so $\Delta q = 0.02\,q_0 = 1$. Compute $MR_0 = 200 \cdot 0.6 \cdot 50^{-0.4} \approx 25.095$. Thus

$$\Delta\pi \approx (25.095 - 20) \cdot 1 = \boxed{\approx 5.10}.$$

22. **Subsidy s in linear market.**
$Q_d = 300 - 3P$, $Q_s = -30 + 2(P + s) = -30 + 2P + 2s$. Equate:

$$-30 + 2P + 2s = 300 - 3P \Rightarrow 5P = 330 - 2s \Rightarrow \boxed{P(s) = 66 - 0.4s},$$

$$Q(s) = 300 - 3P = 300 - 3(66 - 0.4s) = \boxed{102 + 1.2s}.$$

Hence $\boxed{\mathrm{d}P/\mathrm{d}s = -0.4}$ (consumer price falls as subsidy rises), $\boxed{\mathrm{d}Q/\mathrm{d}s = 1.2}$ (quantity increases).

23. **Implicit elasticity from $F(Q, P) = \ln Q + \phi Q - \ln P = 0$ ($\phi > 0$).**
Compute partials: $F_Q = 1/Q + \phi$, $F_P = -1/P$. By implicit differentiation:

$$\left(\frac{1}{Q} + \phi\right)\mathrm{d}Q - \frac{1}{P}\mathrm{d}P = 0 \ \Rightarrow\ \frac{\mathrm{d}Q}{\mathrm{d}P} = \frac{1/P}{1/Q + \phi} = \frac{Q/P}{1 + \phi Q}.$$

Elasticity $\varepsilon = \dfrac{\mathrm{d}Q}{\mathrm{d}P}\dfrac{P}{Q} = \boxed{\dfrac{1}{1 + \phi Q}}$ (positive here because F implies an increasing Q–P relation).